RC Car Drivetrain Engineering Analysis

1/8 Scale Custom Build — Drivetrain Validation Report

Gear Stress · Shaft Analysis · Interference Fits · Top Speed Validation

Motor	Spektrum Firma 1900 KV Brushless
ESC	Spektrum Firma 130 A Smart
Battery	HOOVO 4S LiPo, 14.8 V, 6500 mAh
Wheel Diameter	105 mm
Net Gear Reduction	4:1 (two stages of 2:1)
Scale	1/8

Document Status: **Draft** — Differential calculations pending

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Top Speed Validation

System Parameters

Parameter	Value	Notes
Motor KV Rating	1900 KV	Spektrum Firma
Battery Voltage (nom.)	14.8 V	4S LiPo
Battery Voltage (full)	16.8 V	4.2 V/cell
Net Gear Reduction	4:1	Two 2:1 stages
Wheel Diameter	105 mm	
Wheel Circumference	329.9 mm	$\pi \times 105$

Calculation

Motor RPM at nominal voltage:

$$n_{\text{motor}} = \text{KV} \times V = 1900 \times 14.8 = 28{,}120 \text{ RPM} \tag{1}$$

Wheel RPM after 4:1 reduction:

$$n_{\text{wheel}} = \frac{28{,}120}{4} = 7{,}030 \text{ RPM} \tag{2}$$

Top speed:

$$v = n_{\text{wheel}} \times C_{\text{wheel}} = 7{,}030 \times 0.3299 \times \frac{60}{1000} = \mathbf{139} \text{ km/h} \tag{3}$$

At full charge (16.8 V): $v_{\max} = \mathbf{158}$ km/h

PASS Theoretical top speed 139–158 km/h, well above the 100 km/h target.

Stage 1 Gear Reduction

Gear Specifications

Parameter	Pinion (12T)	Gear (24T)
Module	1 mm	1 mm
Teeth	12	24
Pitch Diameter	12 mm	24 mm
Bore	5 mm	8 mm
Face Width	10 mm	10 mm
Material	Hardened Steel	
Pressure Angle	20°	
Ratio	2:1	

Motor Torque

Maximum motor torque at peak ESC current (130 A):

$$P = V \times I = 14.8 \times 130 = 1924 \text{ W} \tag{4}$$

$$\omega = \frac{2\pi \times 28{,}120}{60} = 2947 \text{ rad/s} \tag{5}$$

$$T_{\text{motor}} = \frac{P}{\omega} = \frac{1924}{2947} = \mathbf{0.653\ N{\cdot}m} \tag{6}$$

Lewis Bending Stress

Tangential force at the pinion pitch circle ($r = 6$ mm):

$$W_t = \frac{T}{r} = \frac{0.653}{0.006} = 108.8 \text{ N} \tag{7}$$

Lewis form factors at 20° pressure angle: $Y_{12T} = 0.245$, $Y_{24T} = 0.337$. Pinion governs (lower Y).

$$\sigma = \frac{W_t}{F \cdot m \cdot Y} = \frac{108.8}{8 \times 1 \times 0.245} = \mathbf{55.5} \text{ MPa} \tag{8}$$

Allowable bending stress for hardened steel: $\sigma_{\text{allow}} = 3000$ MPa

$$SF_{\text{bending}} = \frac{3000}{55.5} = \mathbf{54.05} \tag{9}$$

Hertzian Contact Stress

Reduced radius of curvature at pitch point:

$$R_1 = 6\sin(20°) = 2.052 \text{ mm} \tag{10}$$

$$R_2 = 12\sin(20°) = 4.104 \text{ mm} \tag{11}$$

$$R' = \frac{R_1 R_2}{R_1 + R_2} = 1.368 \text{ mm} \tag{12}$$

Reduced elastic modulus ($E = 200$ GPa, $\nu = 0.3$):

$$E' = \frac{E}{1-\nu^2} = 219.8 \text{ GPa} \tag{13}$$

Normal force: $W_n = W_t/\cos(20°) = 115.8$ N

$$\sigma_H = \sqrt{\frac{W_n \cdot E'}{\pi \cdot F \cdot R'}} = \sqrt{\frac{115.8 \times 219.8 \times 10^9}{\pi \times 0.008 \times 0.001368}} = \mathbf{860.41} \text{ MPa} \tag{14}$$

Allowable surface fatigue stress for hardened steel: $\sigma_{H,\text{allow}} = 3500$ MPa

$$SF_{\text{contact}} = \frac{3500}{860.41} = \mathbf{4.07} \tag{15}$$

Stage 1 Summary

Check	Actual	Allowable	SF
Lewis Bending	44.4 MPa	300 MPa	**54.05**
Hertzian Contact	243.7 MPa	700 MPa	**4.07**

PASS Both bending and contact stress checks pass with comfortable margins.

Intermediate Shaft Analysis

Shaft Specifications

Parameter	Value
Diameter	8 mm
Length	75 mm
Material	304 Stainless Steel
Support	Two bearings, 10 mm inboard of each gear
Bearing Span	35 mm
Overhang (Driven)	30 mm
Overhang (Driving)	10 mm

Corrected Endurance Limit — Marin Factors

The uncorrected endurance limit is estimated as:

$$S'_e = 0.5 \times S_{ut} = 0.5 \times 515 = 257.5 \text{ MPa} \tag{16}$$

The corrected endurance limit applies Marin modification factors:

$$S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot S'_e \tag{17}$$

Factor	Name	Value	Basis
k_a	Surface finish	0.72	Machined surface, 304 SS
k_b	Size	0.85	$d = 8$ mm < 51 mm
k_c	Load type	1.00	Bending
k_d	Temperature	1.00	Ambient operating conditions
k_e	Reliability	0.87	99% reliability target
Combined $k = k_a k_b k_c k_d k_e$		0.533	

$$S_e = 0.533 \times 257.5 = \mathbf{137} \text{ MPa} \approx 130 \text{ MPa} \quad \text{(conservative)} \tag{18}$$

Material Properties — 304 Stainless Steel

Property	Value
Yield Strength S_y	207 MPa
Ultimate Tensile Strength S_{ut}	515 MPa
Endurance Limit (uncorrected)	258 MPa
Corrected Endurance Limit S_e	130 MPa

Bending Moment

By symmetry: $R_A = R_B = 108.8$ N. Maximum bending moment at each bearing (overhang):

$$M_{\max} = W_t \times a = 108.8 \times 0.010 = \mathbf{1.088\ N{\cdot}m} \tag{19}$$

Section Properties (8 mm solid shaft)

$$I = \frac{\pi d^4}{64} = 2.01 \times 10^{-10}\ \mathrm{m}^4, \qquad J = 4.02 \times 10^{-10}\ \mathrm{m}^4, \qquad c = 4\ \mathrm{mm} \tag{20}$$

Stress Analysis

Torque on intermediate shaft:

$$T = 0.653 \times 2 = 1.306\ N{\cdot}m \tag{21}$$

Bending stress:

$$\sigma_b = \frac{M \cdot c}{I} = \frac{1.088 \times 0.004}{2.01 \times 10^{-10}} = 21.7\ \mathrm{MPa} \tag{22}$$

Torsional shear stress:

$$\tau = \frac{T \cdot c}{J} = \frac{1.306 \times 0.004}{4.02 \times 10^{-10}} = 13.0\ \mathrm{MPa} \tag{23}$$

Von Mises equivalent stress:

$$\sigma_{vm} = \sqrt{\sigma_b^2 + 3\tau^2} = \sqrt{21.7^2 + 3(13.0)^2} = \mathbf{28.5}\ \mathrm{MPa} \tag{24}$$

Fatigue Analysis — DE-Goodman

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{21.7}{130} + \frac{13.0}{515} = 0.167 + 0.025 = \mathbf{0.192} \tag{25}$$

$$SF_{\text{fatigue}} = \frac{1}{0.192} = \mathbf{5.2} \tag{26}$$

Yield Check

$$SF_{\text{yield}} = \frac{S_y}{\sigma_{vm}} = \frac{207}{28.5} = \mathbf{7.3} \tag{27}$$

Shaft Summary

Check	Actual	Limit	SF
Fatigue (Goodman)	0.192	1.0	**5.2**
Yield (Von Mises)	28.5 MPa	207 MPa	**7.3**

PASS Intermediate shaft passes both fatigue and yield. Two-bearing arrangement is critical to this result.

Stage 2 Gear Reduction

Gear Specifications

Parameter	Pinion (16T)	Gear (32T)
Module	2 mm	2 mm
Teeth	16	32
Pitch Diameter	32 mm	64 mm
Face Width	20 mm	20 mm
Material	Min. 27 HRC steel (see §4.5)	
Pressure Angle	20°	
Ratio	2:1	

Forces at Stage 2

Torque at Stage 2 input: $T = 1.306\ N{\cdot}m$

$$W_t = \frac{T}{r} = \frac{1.306}{0.016} = 81.6\ \text{N}, \qquad W_n = \frac{81.6}{\cos(20^\circ)} = 86.8\ \text{N} \tag{28}$$

Lewis Bending Stress

$Y_{16T} = 0.296,\ \ Y_{32T} = 0.353.$ Pinion governs.

$$\sigma = \frac{W_t}{F \cdot m \cdot Y} = \frac{81.6}{20 \times 2 \times 0.296} = \mathbf{6.89}\ \text{MPa}, \qquad SF_{\text{bending}} = \frac{170}{6.89} = \mathbf{24.7} \tag{29}$$

Hertzian Contact Stress

$$R_1 = 16\sin(20^\circ) = 5.472\ \text{mm} \tag{30}$$

$$R_2 = 32\sin(20^\circ) = 10.944\ \text{mm} \tag{31}$$

$$R' = \frac{R_1 R_2}{R_1 + R_2} = 3.648\ \text{mm} \tag{32}$$

$$\sigma_H = \sqrt{\frac{W_n \cdot E'}{\pi \cdot F \cdot R'}} = \sqrt{\frac{86.8 \times 219.8 \times 10^9}{\pi \times 0.020 \times 0.003648}} = \mathbf{288.4}\ \text{MPa} \tag{33}$$

Minimum Material Requirement

Gears are machined from available stock; material is back-calculated from a target $SF = 2.0$:

$$\sigma_{H,\text{allow}} = SF \times \sigma_H = 2.0 \times 288.4 = \mathbf{576.8}\ \text{MPa} \tag{34}$$

Minimum Requirement	Value
Surface Fatigue Strength	$\geq$ 577 MPa
Brinell Hardness	$\geq$ 280 HB
Rockwell Hardness	$\geq$ 27 HRC
Yield Strength	$\geq$ 550 MPa

Suitable machinable steels: 4140 (Q&T), 4340, 1045 heat-treated, 4130 chromoly.
Specify to machinist: minimum 27 HRC surface hardness, 550 MPa yield strength.

Stage 2 Summary

Check	Actual	Allowable	SF
Lewis Bending	6.89 MPa	170 MPa	**24.7**
Hertzian Contact	288.4 MPa	577 MPa (design)	**2.0**

PASS Bending stress is negligible. Contact stress sets the material spec — confirm stock meets $\geq$27 HRC before machining.

Interference Fits — Shrink Fit Validation

Overview

Gears are assembled via liquid nitrogen (−196 °C) shrink fit. Two interfaces are analysed.

Assumptions: $\mu = 0.12$ (steel-on-steel), hub outer diameter $= 3d$, $E = 200$ GPa, $\nu = 0.3$.

Interface	Bore	Torque	Shaft Material
Motor → 12T Pinion	5 mm	0.653 $N \cdot m$	Hardened Steel
Shaft → 24T Gear	8 mm	1.306 $N \cdot m$	304 SS

Required Interface Pressure

$$p_{\text{req}} = \frac{2T}{\pi \mu d^2 L} \tag{35}$$

Fit 1 ($d = 5$ mm, $L = 10$ mm):

$$p_1 = \frac{2 \times 0.653}{\pi \times 0.12 \times (0.005)^2 \times 0.010} = \mathbf{13.84}\ \text{MPa} \tag{36}$$

Fit 2 ($d = 8$ mm, $L = 20$ mm):

$$p_2 = \frac{2 \times 1.306}{\pi \times 0.12 \times (0.008)^2 \times 0.020} = \mathbf{17.13}\ \text{MPa} \tag{37}$$

Required Diametral Interference

With $D = 3d$, the Lamé coefficient evaluates to $K = 1.25 + \nu = 1.55$.

$$\delta = \frac{p \cdot d}{E} \cdot K \tag{38}$$

Fit 1:

$$\delta_1 = \frac{13.84 \times 10^6 \times 0.005}{200 \times 10^9} \times 1.55 = 0.54\ \mu\text{m} \tag{39}$$

Fit 2:

$$\delta_2 = \frac{17.13 \times 10^6 \times 0.008}{200 \times 10^9} \times 1.55 = 1.07\ \mu\text{m} \tag{40}$$

Thermal Contraction via Liquid Nitrogen

$\Delta T \approx 216\,°\text{C}$ (ambient to −196 °C). Diametral contraction of shaft:

$$\Delta d = \alpha \cdot d \cdot \Delta T \tag{41}$$

Fit 1 (hardened steel, $\alpha = 12\ \mu\text{m/m}\,°\text{C}$):

$$\Delta d_1 = 12 \times 10^{-6} \times 5 \times 216 = \mathbf{12.96}\ \mu\text{m} \tag{42}$$

Fit 2 (304 SS, $\alpha = 17.2\ \mu\text{m/m}\,°\text{C}$):

$$\Delta d_2 = 17.2 \times 10^{-6} \times 8 \times 216 = \mathbf{29.72}\ \mu\text{m} \tag{43}$$

Assembly Feasibility

Fit	Required δ	LN2 Clearance	Ratio
5 mm (Fit 1)	0.54 μm	12.96 μm	**24×**
8 mm (Fit 2)	1.07 μm	29.72 μm	**28×**

Recommended Tolerance Class

Applying 4× safety margin on minimum interference, both fits are satisfied by the ISO **H7/p6** tolerance class.

Fit	Min. δ	Recommended δ	ISO Class
5 mm bore	0.54 μm	+2.5 μm	**H7/p6**
8 mm bore	1.07 μm	+4.5 μm	**H7/p6**

PASS Both shrink fits are viable. Specify **H7/p6** for both bores. LN2 provides ample assembly clearance (24×–28× excess).

Differential Gear Analysis

Configuration Overview

The differential is a standard open bevel differential. The Stage 2 output drives the **differential casing**, which carries a cross-shaft (spider shaft) on which two **spider gears** are mounted. These spider gears mesh with two **driveshaft side gears** (one per output axle). Under equal loading (straight-line driving), the spider gears do not rotate on the cross-shaft and torque is split 50/50. Under unequal loading (cornering), the spider gears rotate to allow a speed differential between the two axles.

Gear Specifications

Parameter	Value
Type	Straight bevel, 90°
Module	1 mm
Teeth (all gears)	20T
Ratio	1:1
Face Width	7 mm
Shaft Bore	8 mm (side gears)
Spider Shaft Dia.	7 mm
Material	45# Steel (≈ 1045 medium carbon)
Pressure Angle	20°

Material properties — 45# / 1045 medium carbon steel:

Property	Value
Yield Strength S_y	530 MPa
Ultimate Strength S_{ut}	700 MPa
Allowable Bending Stress	250 MPa
Allowable Surface Fatigue	600 MPa

Torque at the Differential

Stage 2 output torque delivers to the differential casing:

$$T_{\text{casing}} = T_{\text{S2 out}} = 1.306 \times 2 = \mathbf{2.612\ N{\cdot}m} \tag{44}$$

Under equal load, each side gear receives:

$$T_{\text{side}} = \frac{T_{\text{casing}}}{2} = \mathbf{1.306\ N{\cdot}m} \tag{45}$$

Each spider gear transmits half the casing torque (two spider gears share the load):

$$T_{\text{spider}} = \frac{T_{\text{casing}}}{2} = \mathbf{1.306\ N{\cdot}m} \tag{46}$$

Bevel Gear Geometry

For a 1:1 ratio, the pitch cone angle:

$$\delta = \arctan\left(\frac{20}{20}\right) = 45^\circ \tag{47}$$

Outer pitch radius and cone distance:

$$r_o = \frac{m \times T}{2} = \frac{1 \times 20}{2} = 10 \text{ mm} \tag{48}$$

$$L_o = \frac{r_o}{\sin(45^\circ)} = \frac{10}{0.7071} = 14.14 \text{ mm} \tag{49}$$

Mean cone distance and mean pitch radius (analysis point for bevel gears):

$$L_m = L_o - \frac{F}{2} = 14.14 - 3.5 = 10.64 \text{ mm} \tag{50}$$

$$r_m = L_m \times \sin(45^\circ) = 10.64 \times 0.7071 = \mathbf{7.524} \text{ mm} \tag{51}$$

Gear Forces

Tangential force at mean radius (same for spider and side gears at 1:1):

$$W_t = \frac{T_{\text{spider}}}{r_m} = \frac{1.306}{0.007524} = \mathbf{173.6} \text{ N} \tag{52}$$

Additional bevel force components (at $\delta = 45^\circ$):

$$W_r = W_t \cdot \tan(20^\circ) \cdot \cos(45^\circ) = 173.6 \times 0.364 \times 0.7071 = \mathbf{44.7} \text{ N} \tag{53}$$

$$W_a = W_t \cdot \tan(20^\circ) \cdot \sin(45^\circ) = \mathbf{44.7} \text{ N} \tag{54}$$

Note: $W_r = W_a$ at 45° pitch cone angle. Axial thrust on each gear is equal to radial force — important for axle bearing selection.

Lewis Bending Stress — Bevel Modified

Mean module:

$$m_m = m \times \left(1 - \frac{F}{2L_o}\right) = 1 \times \left(1 - \frac{7}{28.28}\right) = 0.7525 \text{ mm} \tag{55}$$

Bevel geometry derating factor:

$$K_b = \frac{1}{1 - F/(2L_o)} = \frac{1}{0.7525} = 1.329 \tag{56}$$

Lewis form factor for 20 teeth at 20° PA: $Y = 0.283$

$$\sigma_b = \frac{W_t}{F \cdot m_m \cdot Y} \times K_b = \frac{173.6}{7 \times 0.7525 \times 0.283} \times 1.329 = \mathbf{154.7} \text{ MPa} \tag{57}$$

$$SF_{\text{bending}} = \frac{250}{154.7} = \mathbf{1.62} \tag{58}$$

Hertzian Contact Stress — Bevel

Reduced modulus: $E' = 219.8$ GPa

Equivalent curvature radii at mean pitch point:

$$R_1 = R_2 = r_m \cdot \sin(20°) = 7.524 \times 0.3420 = 2.573 \text{ mm} \tag{59}$$

$$R' = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{2.573^2}{2 \times 2.573} = \mathbf{1.286} \text{ mm} \tag{60}$$

Normal force: $W_n = W_t / \cos(20°) = 173.6/0.9397 = 184.7$ N

$$\sigma_H = \sqrt{\frac{W_n \cdot E'}{\pi \cdot F \cdot R'}} = \sqrt{\frac{184.7 \times 219.8 \times 10^9}{\pi \times 0.007 \times 0.001286}} = \mathbf{379.0} \text{ MPa} \tag{61}$$

$$SF_{\text{contact}} = \frac{600}{379.0} = \mathbf{1.58} \tag{62}$$

Spider Shaft Bending

The spider shaft is modelled as a simply supported beam of span $\approx 2r_o = 20$ mm, with two spider gear loads symmetric about midspan.

$$M = \frac{W_t \times L}{4} = \frac{173.6 \times 0.020}{4} = 0.868 \; N{\cdot}m \tag{63}$$

For the 7 mm solid shaft:

$$I = \frac{\pi d^4}{64} = \frac{\pi (0.007)^4}{64} = 1.179 \times 10^{-10} \text{ m}^4, \qquad c = 3.5 \text{ mm} \tag{64}$$

$$\sigma = \frac{M \cdot c}{I} = \frac{0.868 \times 0.0035}{1.179 \times 10^{-10}} = \mathbf{25.7} \text{ MPa} \tag{65}$$

$$SF_{\text{spider shaft}} = \frac{S_y}{\sigma} = \frac{530}{25.7} = \mathbf{20.6} \tag{66}$$

Differential Summary

Check	Actual	Allowable	SF
Bevel Bending (Lewis mod.)	154.7 MPa	250 MPa	**1.62**
Hertzian Contact	379.0 MPa	600 MPa	**1.58**
Spider Shaft Bending	25.7 MPa	530 MPa	**20.6**

MARGINAL Both gear checks sit at SF ≈ 1.6, below the 2.0 design target. Three mitigating factors apply:

1. Spider gears only experience load differentials *during cornering* — straight-line driving produces no relative rotation and reduces effective cyclic stress.
2. The 45# steel listing reflects minimum properties. If the supplier applies surface hardening (common practice), allowable stresses increase to ~300 MPa bending / 700 MPa contact, raising both SFs above 1.9.
3. Face width and module are fixed by the off-the-shelf part — no geometric im-

provement is possible without changing the gear set.

Recommendation: Confirm with supplier whether gears are surface hardened. If not, this is the most likely failure point in the drivetrain under sustained high-load cornering.

Differential Casing & Assembly Analysis

The differential casing is aluminium. Three structural interfaces are analysed: (1) the ring gear bolted and friction-fitted to the casing, (2) the joint between the two casing halves, and (3) the vertical spider pin bearing on the aluminium casing walls. All bolt tightening torques include a **4× safety factor** over the minimum required.

Casing material — 6061-T6 Aluminium:

Property	Value
Yield Strength S_y	276 MPa
Allowable Bearing Stress	276 MPa

Ring Gear to Differential Casing — Friction Fit (M5 Bolts)

Geometry & assumptions:

- Input torque: $T = 2.6135\ N{\cdot}m$
- Friction radius: $r = 11$ mm
- Friction coefficient: $\mu = 0.2$ (aluminium on steel)
- Bolt spec: M5 (12.9 grade); max preload $F_{\text{bolt,max}} = 9700$ N
- Number of bolts: 4

Required Clamping Force

$$T = \mu \cdot F_{\text{clamp}} \cdot r \implies F_{\text{clamp}} = \frac{T}{\mu \cdot r} = \frac{2.6135}{0.2 \times 0.011} = \mathbf{1187.95}\ \text{N} \tag{67}$$

Force per bolt:

$$F_{\text{clamp,bolt}} = \frac{F_{\text{clamp}}}{4} = \frac{1187.95}{4} = 296.99\ \text{N} \ll 9700\ \text{N} \quad \checkmark \tag{68}$$

Required Tightening Torque (with 4× SF)

Minimum tightening torque (using $K = 0.2$ nut factor):

$$T_{\text{tighten,min}} = K \cdot F_{\text{clamp,bolt}} \cdot d = 0.2 \times 296.99 \times 0.005 = 0.297\ N{\cdot}m \tag{69}$$

Applying 4× safety factor:

$$\boxed{T_{\text{tighten}} = 4 \times 0.297 = \mathbf{1.188\ N{\cdot}m}} \tag{70}$$

This remains well within the M5 12.9 bolt capacity. Torque to this value at assembly.

If Friction Fails — Bolt Shear (Ring Gear / Casing)

Shear force per bolt (4 bolts at $r = 11$ mm):

$$F_{\text{shear}} = \frac{T}{n \cdot r} = \frac{2.6135}{4 \times 0.011} = \mathbf{59.40}\ \text{N} \tag{71}$$

Shear stress in M5 bolt ($d = 5$ mm):

$$\sigma_{\text{shear}} = \frac{F_{\text{shear}}}{\pi/4 \cdot d^2} = \frac{59.40}{\pi/4 \times 5^2} = \mathbf{3.03}\ \text{MPa} \ll 600\ \text{MPa} \quad \checkmark \tag{72}$$

Bearing stress on aluminium casing ($t = 4$ mm wall):

$$\sigma_{\text{bearing}} = \frac{F_{\text{shear}}}{d \cdot t} = \frac{59.40}{5 \times 4} = \mathbf{2.97}\ \text{MPa} \ll 276\ \text{MPa} \quad \checkmark \tag{73}$$

PASS Ring gear / casing interface passes on both friction and fallback bolt shear. Torque M5 bolts to **1.19 N·m** (4× SF applied).

Torque Between Two Halves of Differential Casing — Friction Fit (M2 Bolts)

Geometry & assumptions:

- Input torque: $T = 2.6135\ N{\cdot}m$
- Friction radius: $r = 18.5$ mm
- Friction coefficient: $\mu = 0.6$ (aluminium on aluminium)
- Bolt spec: M2; max preload $F_{\text{bolt,max}} = 1000$ N
- Number of bolts: 8

Required Clamping Force

$$F_{\text{clamp}} = \frac{T}{\mu \cdot r} = \frac{2.6135}{0.6 \times 0.0185} = \mathbf{233.14}\ \text{N} \tag{74}$$

Force per bolt:

$$F_{\text{clamp,bolt}} = \frac{233.14}{8} = 29.14\ \text{N} \ll 1000\ \text{N} \quad \checkmark \tag{75}$$

Required Tightening Torque (with 4× SF)

Minimum tightening torque ($K = 0.2$, $d = 2$ mm):

$$T_{\text{tighten,min}} = K \cdot F_{\text{clamp,bolt}} \cdot d = 0.2 \times 29.14 \times 0.002 = 0.01166\ N{\cdot}m \tag{76}$$

Applying 4× safety factor:

$$\boxed{T_{\text{tighten}} = 4 \times 0.01166 = \mathbf{0.0466\ N{\cdot}m}} \tag{77}$$

If Friction Fails — Bolt Shear (Casing Halves)

Shear force per bolt (8 bolts at $r = 18.5$ mm):

$$F_{\text{shear}} = \frac{T}{n \cdot r} = \frac{2.6135}{8 \times 0.0185} = \mathbf{17.66}\ \text{N} \tag{78}$$

Shear stress in M2 bolt ($d = 2$ mm):

$$\sigma_{\text{shear}} = \frac{F_{\text{shear}}}{\pi/4 \cdot d^2} = \frac{17.66}{\pi/4 \times 4} = \mathbf{5.62}\ \text{MPa} \ll 600\ \text{MPa} \quad \checkmark \tag{79}$$

Bearing stress on aluminium casing ($t = 2$ mm wall):

$$\sigma_{\text{bearing}} = \frac{F_{\text{shear}}}{d \cdot t} = \frac{17.66}{4 \times 2} = \mathbf{2.21}\ \text{MPa} \ll 276\ \text{MPa} \quad \checkmark \tag{80}$$

PASS Casing half joint passes on both friction and fallback bolt shear. Torque M2 bolts to **0.047 N·m** (4× SF applied).

Vertical Spider Pin — Bearing Stress on Aluminium Casing

The spider pin (cross-shaft, ∅7 mm) transmits the full differential reaction force into the aluminium casing walls. The pin bears against two casing walls of thickness 4 mm each (total bearing length = 2 × 4 = 8 mm), and the pin sits at a radius of $r = 22$ mm from the casing centreline.

Pin Bearing Force

$$F_{\text{pin}} = \frac{T}{r} = \frac{2.6135}{0.022} = \mathbf{118.80}\ \text{N} \tag{81}$$

Bearing Stress on Casing

Contact area per side: $A = d_{\text{pin}} \times t_{\text{wall}} = 7 \times 4 = 28\ \text{mm}^2$. Two walls share the load:

$$\sigma_{\text{bearing}} = \frac{F_{\text{pin}}}{2 \times d_{\text{pin}} \times t_{\text{wall}}} = \frac{118.80}{2 \times 7 \times 4} = \mathbf{2.12}\ \text{MPa} \ll 276\ \text{MPa} \quad \checkmark \tag{82}$$

$$SF_{\text{pin bearing}} = \frac{276}{2.12} = \mathbf{130} \tag{83}$$

PASS Spider pin bearing stress on aluminium casing is negligible. SF = 130.

Casing Assembly Summary

Check	Actual	Allowable	Status
Ring gear friction (M5, 4× SF)	$T = 1.19\ N{\cdot}m$	< 9700 N preload	**PASS**
Ring gear bolt shear (fallback)	3.03 MPa	600 MPa	**PASS**
Ring gear bearing on Al (fallback)	2.97 MPa	276 MPa	**PASS**
Casing half friction (M2, 4× SF)	$T = 0.047\ N{\cdot}m$	< 1000 N preload	**PASS**
Casing half bolt shear (fallback)	5.62 MPa	600 MPa	**PASS**
Casing half bearing on Al (fallback)	2.21 MPa	276 MPa	**PASS**
Spider pin bearing on Al casing	2.12 MPa	276 MPa	**PASS**

Assembly torque values (with 4× SF):

- Ring gear → casing (M5 × 4): **1.19 N·m per bolt**
- Casing halves (M2 × 8): **0.047 N·m per bolt**

All casing interfaces pass with very large margins. The aluminium casing is not the

limiting component.

Overall Summary

Analysis	Component	Result	Status
Top Speed	Full drivetrain	139 km/h	**PASS**
Lewis Bending	Stage 1 gears	SF = 6.75	**PASS**
Hertzian Contact	Stage 1 gears	SF = 2.87	**PASS**
Fatigue (Goodman)	Int. shaft	SF = 5.2	**PASS**
Yield (Von Mises)	Int. shaft	SF = 7.3	**PASS**
Lewis Bending	Stage 2 gears	SF = 24.7	**PASS**
Hertzian Contact	Stage 2 gears	SF = 2.0 (design)	**PASS**[†]
Shrink Fit — 5 mm	Motor / Pinion	24× margin	**PASS**
Shrink Fit — 8 mm	Shaft / Gear	28× margin	**PASS**
Bevel Bending	Differential gears	SF = 1.62	**MARGINAL**[‡]
Hertzian Contact	Differential gears	SF = 1.58	**MARGINAL**[‡]
Spider Shaft	Differential	SF = 20.6	**PASS**
Ring gear friction fit	Diff casing (M5)	$T = 1.19\ N{\cdot}m$	**PASS**
Ring gear bolt shear	Diff casing fallback	3.03 MPa	**PASS**
Ring gear Al bearing	Diff casing fallback	2.97 MPa	**PASS**
Casing half friction fit	Diff casing (M2)	$T = 0.047\ N{\cdot}m$	**PASS**
Casing half bolt shear	Diff casing fallback	5.62 MPa	**PASS**
Casing half Al bearing	Diff casing fallback	2.21 MPa	**PASS**
Spider pin Al bearing	Diff casing	SF = 130	**PASS**

[†]Stage 2 contact stress drives the material specification. Stock must meet ≥27 HRC / 550 MPa yield before fabrication.

[‡]Differential gear SFs are below the 2.0 target but are mitigated by intermittent loading (cornering only) and likely supplier surface hardening. Confirm heat treatment with supplier before accepting parts.

Pending analyses:

- Stage 2 shaft stress and fatigue

All other checks complete. Design is viable for the 100 km/h target. The differential bevel gear set is the most stress-critical component; all casing interfaces pass with very large margins.